SANGER: CLONED AMPLICONS

one clone is one molecule, read once



each clone's own mutations

529-bp window that the clones read cleanly;
447 chromatograms from the six mice of Figure 5

● mutation ● sequencing error

NANOPORE: CLONING-FREE AMPLICONS

every read spans the 1,230-bp amplicon



529-bp window

Eight wild-type reads over the whole amplicon, with every mutation called in them; errors drawn at the rate the controls show, 1 base in 700, pale below Q 20

From the Sanger clones here

Each molecule's own mutations
the mutation load of every clone, the share of clones mutated, and frequency as published: unique mutations over the bases each clone read

Bookkeeping checks
copied traces, clones filed under two mice, mutated clones filed as controls, mislabeled tubes (Figure 5E)

Error reported, not subtracted
the AID-null clones' error rate (about 20 per 10^5 bases at Q ≥ 20 in both published studies), and how many of each animal's calls it explains

From both, and they agree

Wild type against AID-null
Sanger 38-fold; nanopore 141-fold, with controls without calls counted at one event

Substitution spectrum
r = 0.92 (95% CI 0.74–0.98), slope 0.97

Position along the window
r = 0.94 in 25-bp bins

AID hotspot use
G:C at WRC/GYW 58% and 54% (20% of sites);
A:T at WA/TW 70% and 69% (50% of sites)

Not comparable: absolute density
a clone is read once; nanopore counts unique mutations among 500 reads. Compare densities within a platform, not between.

From the nanopore reads here

The whole amplicon
positions 90–1,230: 229-fold over controls; 69% of mutations in the first 450 bp, falling 7-fold to the last 250 bp (Figure 6A)

Clones and mutation trees
clones of each mouse, and their mutations ordered into trees (Figures 6 and S11)

Calls at an estimated false-discovery rate
each candidate tested against the controls, the threshold set by them as decoys; single-read mutations not called, as one background-subtracted estimate